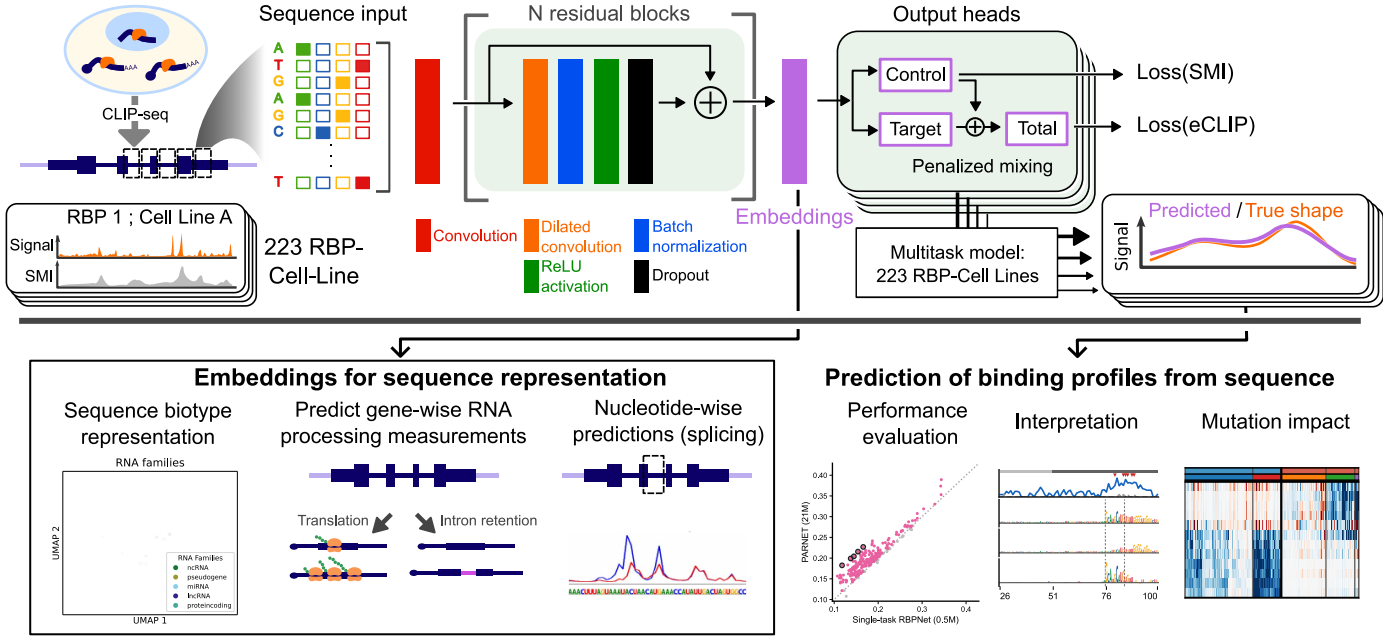

CLIP-seq
Sequence input
A
T
G
A
G
C
T
N residual blocks
Output heads
Control
Target
Total
Loss(SMI)
Loss(eCLIP)
Penalized mixing
Embeddings
RBP 1 ; Cell Line A
Signal
SMI
223 RBP-Cell-Line
Convolution
Dilated convolution
Batch normalization
ReLU activation
Dropout
Multitask model: 223 RBP-Cell Lines
Predicted / True shape
Signal
Embeddings for sequence representation
Sequence biotype representation
RNA families
UMAP 2
UMAP 1
RNA Families
ncRNA
pseudogene
miRNA
lncRNA
proteincoding
Predict gene-wise RNA processing measurements
Translation
Intron retention
Nucleotide-wise predictions (splicing)
Prediction of binding profiles from sequence
Performance evaluation
PARNET (21M)
Single-task RBPNet (0.5M)
Interpretation
Mutation impact